

Unambiguous Quantum State Discrimination with Error Tolerance

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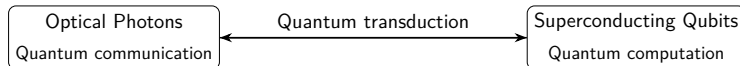
- Coherent states
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Introduction

Motivation: Quantum transduction

Meesala et al. (2024)

(Future, large-scale quantum computers and quantum networks would) leverage the complementary strengths of optical photons and superconducting microwave circuits to reliably communicate and process quantum information



Why non-orthogonal quantum states are important

- BB84 - The first quantum key distribution protocol
- Inspired by non-orthogonal quantum states
- Still the mainstream protocol with countermeasures for attacks in mind

QUANTUM TRANSMISSION															
Alice's random bits	0	1	1	0	1	1	0	0	1	0	1	1	0	0	1
Random sending bases	D	R	D	R	R	R	R	R	D	D	R	D	D	D	R
Photons Alice sends	$\nearrow$	$\downarrow$	$\nwarrow$	$\leftrightarrow$	$\uparrow$	$\downarrow$	$\leftrightarrow$	$\leftrightarrow$	$\nwarrow$	$\nearrow$	$\downarrow$	$\nwarrow$	$\nearrow$	$\downarrow$	$\uparrow$
Random receiving bases	R	D	D	R	R	D	D	R	D	R	D	D	D	D	R
Bits as received by Bob	1		1		1	0	0	0		1	1	1		0	1
PUBLIC DISCUSSION															
Bob reports bases of received bits	R		D		R	D	D	R		R	D	D		D	R
Alice says which bases were correct			OK		OK			OK				OK		OK	OK
Presumably shared information (if no eavesdrop)			1		1			0				1		0	1
Bob reveals some key bits at random					1									0	
Alice confirms them					OK									OK	
OUTCOME															
Remaining shared secret bits			1					0				1			1

Figure: The process of the BB84 protocol

BB84 - Optical experiments

- Doing research for QKD involves expensive infrastructure, precise hardware, and domain knowledge in quantum optics

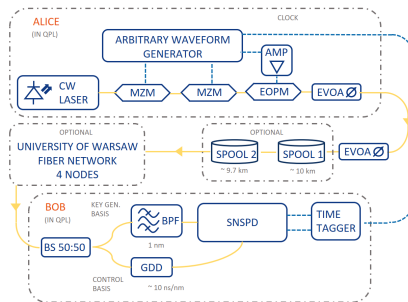


Figure: The experiment architecture of decoy-state BB84

Our motivation

- Easier access of quantum computers may enable researchers to bypass the difficulty of experimental demonstration
- The performance of the quantum circuits may serve as a benchmark from the view of quantum circuit optimization and quantum hardware design
- If lucky, quantum circuits for quantum state discrimination may inspire new research topics

Our contributions

- A confidence-aware formulation for noisy quantum state discrimination
- A distribution-aware formulation for noisy quantum state discrimination
- A quantum circuit synthesis flow for quantum state discrimination

Preliminaries

Preliminaries

Basics of Quantum Computation

Dirac notation

In quantum mechanics, we usually use the Dirac notation (bra-ket notation) to denote quantum states.

- A ket in $\mathbb{C}^n$: $|\psi\rangle \equiv \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ (state vector)
- A bra $\langle\psi| \equiv |\psi\rangle^\dagger = (x_1^* \ x_2^* \ \cdots \ x_n^*)$
- Usually, $\sum_{i=1}^n |x_i|^2 = 1$
- For example, the state vector of a qubit is $|\psi\rangle = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$,
where $|x_1|^2 + |x_2|^2 = 1$, $x_1, x_2 \in \mathbb{C}$

Describing composite systems

Nielsen and Chuang (2011)

Postulate 4 (of quantum mechanics):

The state space of a composite physical system is the tensor product of the state spaces of the component physical systems. Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$.

Density matrix

However, when the system we concern (denoted as A) is not closed (dependent of the environment, denoted as B), the state vector alone is not enough to describe the system. We can expand the state vector to be a density matrix.

- Density matrix for $|\psi\rangle$: $\rho \equiv |\psi\rangle \langle\psi|$
- Important property: $\text{tr } \rho = 1$
- For independent systems: $\rho^{AB} = \rho^A \otimes \rho^B$
- For dependent systems: $\rho^{AB} \neq \rho^A \otimes \rho^B$

We can think of the system's density matrix as the partial trace of the density matrix of the system and the environment together (AB).

- $\rho^A = \text{tr}_B \rho^{AB}$

Quantum measurements

Nielsen and Chuang (2011)

Postulate 3 (of quantum mechanics):

Quantum measurements are described by a collection M_m of measurement operators. These are operators acting on the state space of the system being measured. The index m refers to the measurement outcomes that may occur in the experiment. If the state of the quantum system is $|\psi\rangle$ immediately before the measurement then the probability that result m occurs is given by

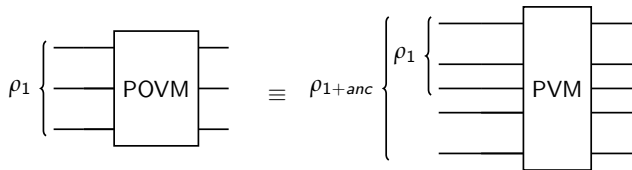
$$p(m) = \langle \psi | M_m^\dagger M_m | \psi \rangle . \quad (1)$$

where $\sum_m M_m^\dagger M_m = I$.

Positive Operator-Valued Measure (POVM)

We define a POVM element $\Pi_m \equiv M_m^\dagger M_m$. A POVM is the complete set of POVM elements $\{\Pi_m\}$ such that

- Π_m is a positive operator
- $\sum_m \Pi_m = I$
- $p(m) = \langle \psi | \Pi_m | \psi \rangle$



POVMs are necessary to describe the effect on a subsystem of a projective measurement performed on a larger system.

Preliminaries

Basics of Quantum State Discrimination (QSD)

Problem description (1/2)

Let $S_{|\psi\rangle} \equiv \{|\psi_1\rangle, |\psi_2\rangle, \dots, |\psi_m\rangle\}$ is a set of non-orthogonal quantum states, known to Alice and Bob before communication.

Let $S_{prior} \equiv \{p_1, p_2, \dots, p_m\}$ be the set of prior probabilities corresponding to each state in $S_{|\psi\rangle}$. The prior probability p_i can be understood as the assumed ratio of Alice preparing and sending the state $|\psi_i\rangle$ to Bob.

For example, $m = 2$, $|\psi_1\rangle = |0\rangle$ and $|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$.

$\langle\psi_1|\psi_2\rangle \neq 0$. $S_{prior} = \{p_1, p_2\} = \{0.5, 0.5\}$.

Problem description (2/2)

Given the predefined set of states $S_{|\psi\rangle}$ and the prior probabilities S_{prior} , the quantum state discrimination problem discusses different ways to measure and identify the quantum state prepared by Alice.

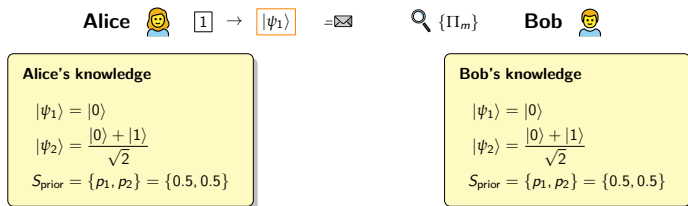


Figure: Alice selects a quantum state and sends it to Bob, who performs a measurement.

Projective measurements

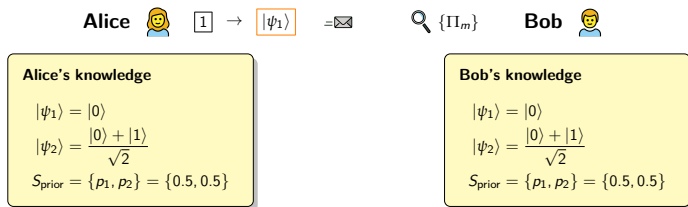
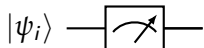


Figure: Alice selects a quantum state and sends it to Bob, who performs a measurement.



If Bob measures with $\Pi_1 = |0\rangle\langle 0|$, $\Pi_2 = |1\rangle\langle 1|$, $P_{\text{succ}} = p_1 \langle \psi_1 | \Pi_1 | \psi_1 \rangle + p_2 \langle \psi_2 | \Pi_2 | \psi_2 \rangle = 0.5 \times 1 + 0.5 \times 0.5 = 0.75$

Probability matrix

We will show the probabilities of each state i and measurement outcome m in the following "probability matrices".

$$p(m, i) \equiv p(\text{Bob measures } m, \text{Alice prepares } i)$$

$$p(m|i) \equiv p(\text{Bob measures } m | \text{Alice prepares } i)$$

	Π_1	Π_2
$ \psi_1\rangle$	$p(1, 1)$	$p(2, 1)$
$ \psi_2\rangle$	$p(1, 2)$	$p(2, 2)$

Table: Probability matrix $p(m, i) = p_i \langle \psi_i | \Pi_m | \psi_i \rangle$

Minimum error discrimination (MED)

If Bob measures with $\Pi_1 = |0\rangle\langle 0|$, $\Pi_2 = |1\rangle\langle 1|$, $P_{succ} = 0.75$.

Can we do better?

Minimum error discrimination aims to find a POVM with m elements such that the overall probability of success is maximized.

	Π_1	Π_2
$ \psi_1\rangle$	correct	error
$ \psi_2\rangle$	error	correct

Table: Probability matrix $p(m, i) = p_i \langle \psi_i | \Pi_m | \psi_i \rangle$

Minimum error discrimination for two states

For $m = 2$, the optimal probability of success P_{succ} can be analytically solved.

$$\sum_{i=1}^m \Pi_i = \Pi_1 + \Pi_2 = I \quad (2)$$

$$P_{succ} = \sum_{i=1}^m p_i \langle \psi_i | \Pi_i | \psi_i \rangle \quad (3)$$

$$= p_1 \langle \psi_1 | \Pi_1 | \psi_1 \rangle + p_2 \langle \psi_2 | \Pi_2 | \psi_2 \rangle \quad (4)$$

$$= p_1 \langle \psi_1 | \Pi_1 | \psi_1 \rangle + p_2 \langle \psi_2 | (I - \Pi_1) | \psi_2 \rangle \quad (5)$$

$$= p_2 + \text{Tr} [(p_1 |\psi_1\rangle \langle \psi_1| - p_2 |\psi_2\rangle \langle \psi_2|) \Pi_1] \quad (6)$$

For our example, the probability of success for MED is

$$P_{succ} = (2 + \sqrt{2})/4 = 0.85355 \dots > 0.75 \quad (7)$$

(Go to Frame 72 for details)

Unambiguous quantum state discrimination (UQSD)

Continue with $|\psi_1\rangle = |0\rangle$ and $|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$,

$S_{prior} = \{p_1, p_2\} = \{0.5, 0.5\}$.

Below is an example of a POVM that performs UQSD.

$$\Pi_1 \equiv \Pi_1^\perp \equiv \frac{\sqrt{2}}{1 + \sqrt{2}} \frac{(|0\rangle - |1\rangle)(\langle 0| - \langle 1|)}{2} \quad (8)$$

$$\Pi_2 \equiv \Pi_2^\perp \equiv \frac{\sqrt{2}}{1 + \sqrt{2}} |1\rangle \langle 1| \quad (9)$$

$$\Pi_3 \equiv \Pi_? \equiv I - \Pi_1 - \Pi_2 \quad (10)$$

Probability matrices for the UQSD example

	$\Pi_1(\Pi_1^\perp)$	$\Pi_2(\Pi_2^\perp)$	$\Pi_3(\Pi_?)$
$ \psi_1\rangle$	0.293	0	0.707
$ \psi_2\rangle$	0	0.293	0.707

Table: Probability matrix $p(m|i) = \langle \psi_i | \Pi_m | \psi_i \rangle$

	$\Pi_1(\Pi_1^\perp)$	$\Pi_2(\Pi_2^\perp)$	$\Pi_3(\Pi_?)$
$ \psi_1\rangle$	0.146	0	0.354
$ \psi_2\rangle$	0	0.146	0.354

Table: Probability matrix $p(m, i) = p_i \langle \psi_i | \Pi_m | \psi_i \rangle$

Preliminaries

Optimal POVM for UQSD

Previous work – Optimal POVM for UQSD

Eldar (2003b)

Although the two-state problem is well developed, the problem of unambiguous discrimination between multiple quantum states has received considerably less attention.

This work (we abbrev. "OP-UQSD") proposed a semidefinite programming approach for unambiguous discrimination between multiple quantum states. We will quickly go through its problem formulation.

Problem formulation for Optimal UQSD

The finite set of quantum states $S_{|\psi\rangle}$ are linearly-independent and non-orthogonal.

For example,

$$|\psi_1\rangle = \frac{1}{\sqrt{1+a_1^2}} (|00\rangle + a_1 |11\rangle), \quad (11)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{1+a_2^2}} (|01\rangle + a_2 |11\rangle), \quad (12)$$

$$|\psi_3\rangle = \frac{1}{\sqrt{1+a_3^2}} (|10\rangle + a_3 |11\rangle) \quad (13)$$

where $a_i \in \mathbb{R}, 0 < a_i < 1$ for $i \in \{1, 2, 3\}$.

For example

$$|\psi_1\rangle = \frac{1}{\sqrt{1+a_1^2}} (|00\rangle + a_1 |11\rangle),$$

$$|\psi_2\rangle = \frac{1}{\sqrt{1+a_2^2}} (|01\rangle + a_2 |11\rangle),$$

$$|\psi_3\rangle = \frac{1}{\sqrt{1+a_3^2}} (|10\rangle + a_3 |11\rangle)$$

For example, given $a_1 = 0.2$, $a_2 = 0.5$, and $a_3 = 0.7$. We can express the states in state vectors, where the floating point numbers are rounded to 4 digits.

$$|\psi_1\rangle = (0.9806 + 0.j \quad 0. + 0.j \quad 0. + 0.j \quad 0.1961 + 0.j)^T$$

$$|\psi_2\rangle = (0. + 0.j \quad 0.8944 + 0.j \quad 0. + 0.j \quad 0.4472 + 0.j)^T$$

$$|\psi_3\rangle = (0. + 0.j \quad 0. + 0.j \quad 0.8192 + 0.j \quad 0.5735 + 0.j)^T$$

Reciprocal states

- The reciprocal states $|\tilde{\phi}_i\rangle$ are unique vectors such that

$$\langle \tilde{\phi}_i | \psi_k \rangle = \delta_{ik}, \forall 1 \leq i, k \leq m. \quad (14)$$

- By stacking the vectors in $S_{|\psi\rangle}(\Psi)$, we can calculate the reciprocal states all at once (Φ).

$$\Psi = \begin{pmatrix} 0.9806 + 0.j & 0. + 0.j & 0. + 0.j & 0.1961 + 0.j \\ 0. + 0.j & 0.8944 + 0.j & 0. + 0.j & 0.4472 + 0.j \\ 0. + 0.j & 0. + 0.j & 0.8192 + 0.j & 0.5735 + 0.j \end{pmatrix}^T$$

$$\Phi = \Psi(\Psi^* \Psi)^{-1}$$

$$= \begin{pmatrix} 0.9969 + 0.j & -0.0573 + 0.j & -0.0802 + 0.j & 0.1146 + 0.j \\ -0.0628 + 0.j & 0.961 + 0.j & -0.2198 + 0.j & 0.3141 + 0.j \\ -0.096 + 0.j & -0.24 + 0.j & 0.8846 + 0.j & 0.48 + 0.j \end{pmatrix}^T$$

Semidefinite programming for OP-UQSD

We can formulate a semidefinite programming problem using the reciprocal states calculated in last frame.

$$\Pi_i \equiv c_i |\tilde{\phi}_i\rangle \langle \tilde{\phi}_i|, 1 \leq i \leq m$$

$$\begin{array}{ll} \underset{\vec{c}=\{c_1, \dots, c_m\}}{\text{minimize}} & 1 - \sum_{i=1}^m p_i c_i \end{array} \quad (15)$$

$$\begin{array}{ll} \text{subject to} & \Pi_{\text{?}} = I - \sum_{i=1}^m c_i |\tilde{\phi}_i\rangle \langle \tilde{\phi}_i| \succeq 0 \end{array} \quad (16)$$

	Π_1	Π_2	Π_3	$\Pi_{\text{?}}$
$ \psi_1\rangle$	$p_1 c_1$	0	0	$p_1(1 - c_1)$
$ \psi_2\rangle$	0	$p_2 c_2$	0	$p_2(1 - c_2)$
$ \psi_3\rangle$	0	0	$p_3 c_3$	$p_3(1 - c_3)$

Table: Probability matrix of the example

Result for the example

Let $S_{prior} = \{1/3, 1/3, 1/3\}$, we get $\vec{c} = \{0.9615, 0.8, 0.6711\}$

Error-Tolerant QSD

Problem formulation

We try to discuss quantum state discrimination with the **noise**.

$$\text{Noise } \mathcal{E}(\rho) = (1 - \lambda)\rho + \lambda \text{Tr}[\rho] \frac{I}{2^n}$$

Alice  $|1\rangle \rightarrow |\psi_1\rangle$ =   $\{\Pi_m\}$ Bob 

Alice's knowledge

$$|\psi_1\rangle = |0\rangle$$

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$S_{\text{prior}} = \{p_1, p_2\} = \{0.5, 0.5\}$$

Bob's knowledge

$$|\psi_1\rangle = |0\rangle$$

$$|\psi_2\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$S_{\text{prior}} = \{p_1, p_2\} = \{0.5, 0.5\}$$

Previous work - FRIO

Assume that the input quantum states are **mixed states** instead of pure states.

For minimum error discrimination, the formulation is unchanged.

For unambiguous discrimination, it becomes almost impossible to find a solution.

Eldar (2003a) Fixed rate of inconclusive outcome (FRIO)

	Π_1	Π_2	Π_3	$\Pi_?$
ρ_1				
ρ_2				β
ρ_3				

Table: Probability matrix $p(m, i) = p_i \langle \psi_i | \Pi_m | \psi_i \rangle$

Semidefinite programming for FRIO

	Π_1	Π_2	Π_3	$\Pi_?$
ρ_1				
ρ_2				β
ρ_3				

$$\begin{array}{ll} \text{maximize} & \text{■ } P_D = \sum_{i=1}^m p_i \text{Tr}(\rho_i \Pi_i) \end{array} \quad (17)$$

$$\begin{array}{ll} \text{subject to} & \Pi_i \succeq 0, \quad i = 0, \dots, m \end{array} \quad (18)$$

$$\sum_{i=0}^m \Pi_i = I \quad (19)$$

$$\text{■ } P_{inc} = \sum_{i=1}^m p_i \text{Tr}(\rho_i \Pi_0) \geq \beta \quad (20)$$

Our problem formulation 1 – CrossQD

Inspired by the type-I error and type-II error in hypothesis testing, we derive a new set of constraints.

	Π_1	Π_2
ρ_1	True positive	False negative
ρ_2	False positive	True negative

Table: An analogy to the confusion matrix

	Π_1	Π_2	Π_3	$\Pi_?$
ρ_1		*		
ρ_2	—	+	—	
ρ_3		*		

Semidefinite programming for CrossQD

	Π_1	Π_2	Π_3	$\Pi_?$
ρ_1		*		
ρ_2	—	+	—	
ρ_3		*		

$$\begin{array}{ll} \underset{\{\Pi\}}{\text{maximize}} & P_D = \sum_{i=1}^m p_i \text{Tr}(\rho_i \Pi_i) \end{array} \quad (21)$$

$$\text{subject to} \quad \Pi_i \succeq 0, \quad i = 0, \dots, m \quad (22)$$

$$\sum_{i=0}^m \Pi_i = I \quad (23)$$

$$- \quad P(O_i | \rho_i) = \frac{p_i \text{Tr}(\rho_i \Pi_i)}{\sum_{j=1}^m p_j \text{Tr}(\rho_j \Pi_j)} \geq 1 - \alpha_i, \quad i = 1, \dots, m \quad (24)$$

$$* \quad P(\rho_i | O_i) = \frac{p_i \text{Tr}(\rho_i \Pi_i)}{\sum_{j=1}^m p_j \text{Tr}(\rho_j \Pi_j)} \geq 1 - \beta_i, \quad i = 1, \dots, m \quad (25)$$

Benefits of CrossQD

- We can make sure the resulting POVM keeps the states we want.
- We have more flexibility to tweak the POVM than with FRIO.

Problem formulation 2 – FitQD

Goal: Find a POVM such that the resulting probability distribution is the same as the ideal distribution π , generated by OP-UQSD.

Question: What does the "distribution" mean here?

	Π_1	Π_2	Π_3	$\Pi_4 = \Pi_?$
ρ_1				
ρ_2				
ρ_3				
sum of column	μ_1	μ_2	μ_3	μ_4
OP-UQSD	π_1	π_2	π_3	π_4

Table: Our definition of the distribution that we want to approach

Semidefinite programming for FitQD

	Π_1	Π_2	Π_3	$\Pi_4 = \Pi_?$
ρ_1				
ρ_2				
ρ_3				
sum of column	μ_1	μ_2	μ_3	μ_4
OP-UQSD	π_1	π_2	π_3	π_4

$$\begin{array}{ll} \text{maximize}_{\{\Pi\}} & P_D = \sum_{i=1}^m p_i \operatorname{Tr}(\rho_i \Pi_i) \end{array} \quad (26)$$

$$\text{subject to} \quad \Pi_i \succeq 0, \quad i = 0, \dots, m \quad (27)$$

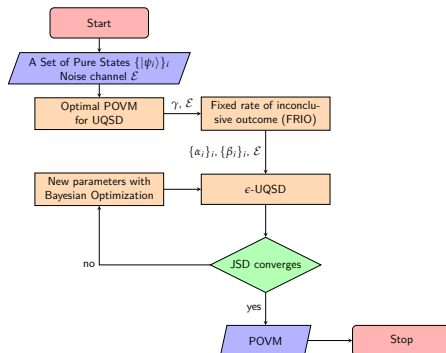
$$\sum_{i=0}^m \Pi_i = I \quad (28)$$

$$\mu_i = \sum_{j=1}^m p_j \operatorname{Tr}(\rho_j \Pi_i) = \pi_i, \quad i = 1, \dots, m+1 \quad (29)$$

Some other attempts

We have tried several approaches before we came up with FitQD.

- 1 Use the POVM from OP-UQSD
- 2 Gradient descent/Bayesian optimization as outer-loop, solving SDP as inner-loop (a mess)



Comparison of our attempts

simple_2 is the set of states mentioned in Frame 28.

Dep. noise level	ideal optimal	FRIO (ideal pinc)	crossQD (0.75*noise)	crossQD-2 (ideal params, ideal pinc)	FitQD (ideal distrib)
1.00E-06	0.000000	0.002606	0.001185	0.000003	0.000000
3.16E-05	0.000000	0.002605	0.002086	0.000003	0.000000
1.00E-05	0.000002	0.002601	0.003583	0.000006	0.000000
3.16E-04	0.000005	0.002558	0.006124	0.000054	0.000000
1.00E-04	0.000015	0.002120	0.009940	0.000202	0.000000
3.16E-03	0.000049	0.000022	0.015885	0.000044	0.000000
1.00E-03	0.000154	0.000068	0.023398	0.000073	0.000000
3.16E-02	0.000487	0.000247	0.034139	0.000094	0.000000
1.00E-02	0.001538	0.001412	0.054889	0.000263	0.000000
3.16E-01	0.004845	0.002258	0.105572	0.001372	0.000000
1.00E-01	0.015152	0.007085	0.226254	0.005255	0.000000

Table: Jensen-Shannon distance comparison (The lower the better) for simple_2 under different noise levels.

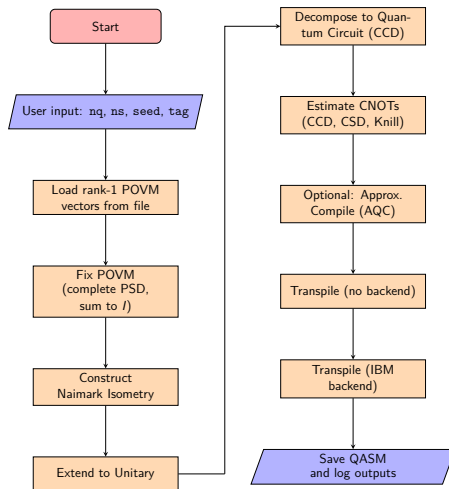
Comparison of different formulations

	OP-UQSD	FRIO	CrossQD	FitQD
Our proposal			✓	✓
Max P_{succ}	✓	✓	✓	✓
Noise aware		✓	✓	✓
Confidence aware			✓	
Distribution aware	-			✓

Table: Comparison of different formulations

Quantum Circuit Synthesis for QSD

Synthesis flow

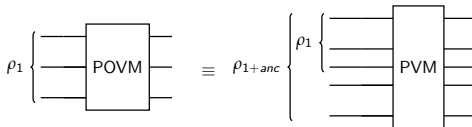


Naimark's dilation theorem (1/5)

Naimark's dilation theorem shows how POVMs can be obtained from PVMs acting on a larger space. (Refer to Frame 15)

Theorem (Naimark's dilation theorem)

If $\{F_i\}_{i=1}^n$ is a POVM acting on a Hilbert space $\mathcal{H}_A$ of dimension d_A , then there exists a PVM $\{\Pi_i\}_{i=1}^n$ acting on a Hilbert space $\mathcal{H}_{A'}$ of dimension $d_{A'}$ and an isometry $V : \mathcal{H}_A \rightarrow \mathcal{H}_{A'}$ such that for all i , $F_i = V^\dagger \Pi_i V$.



Naimark's dilation theorem (2/5)

Theorem (Naimark's dilation theorem)

If $\{F_i\}_{i=1}^n$ is a POVM acting on a Hilbert space $\mathcal{H}_A$ of dimension d_A , then there exists a PVM $\{\Pi_i\}_{i=1}^n$ acting on a Hilbert space $\mathcal{H}_{A'}$ of dimension $d_{A'}$ and an isometry $V : \mathcal{H}_A \rightarrow \mathcal{H}_{A'}$ such that for all i , $F_i = V^\dagger \Pi_i V$.

For the particular case of a **rank-1 POVM**, i.e., when $F_i = |f_i\rangle\langle f_i|$ for some (unnormalized) vectors $|f_i\rangle$, this isometry can be constructed as

$$V = \sum_{i=1}^n |i\rangle_{A'} \langle f_i|_A \quad (30)$$

and the PVM is given simply by $\Pi_i = |i\rangle\langle i|_{A'}$. Note that here $d_{A'} = n$.

Naimark's dilation theorem (3/5)

Theorem (Naimark's dilation theorem)

If $\{F_i\}_{i=1}^n$ is a POVM acting on a Hilbert space $\mathcal{H}_A$ of dimension d_A , then there exists a PVM $\{\Pi_i\}_{i=1}^n$ acting on a Hilbert space $\mathcal{H}_{A'}$ of dimension $d_{A'}$ and an isometry $V : \mathcal{H}_A \rightarrow \mathcal{H}_{A'}$ such that for all i , $F_i = V^\dagger \Pi_i V$.

In the general case, the isometry and PVM can be constructed by defining $\mathcal{H}_{A'} = \mathcal{H}_A \otimes \mathcal{H}_B$, $\Pi_i = \mathbb{I}_A \otimes |i\rangle\langle i|_B$, and

$V = \sum_{i=1}^n \sqrt{F_{iA}} \otimes |i\rangle_B$. Note that here $d_{A'} = nd_A$, so this is a more wasteful construction.

Naimark's dilation theorem (4/5)

- **The probability is transformation-invariant**

In either case, the probability of obtaining outcome i with this PVM, and the state suitably transformed by the isometry, is the same as the probability of obtaining it with the original POVM:

$$\text{Prob}(i) = \text{tr} \left(V \rho_A V^\dagger \Pi_i \right) = \text{tr} \left(\rho_A V^\dagger \Pi_i V \right) = \text{tr}(\rho_A F_i)$$

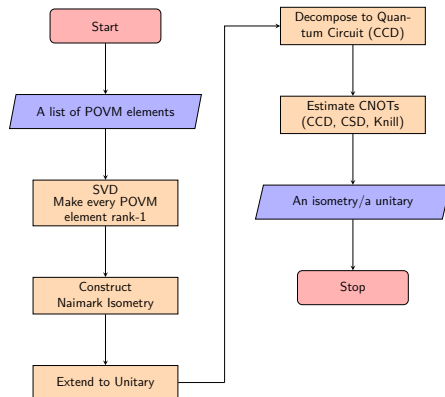
- **A recipe for a physical realisation of the POVM**

We can always do it by extending the isometry V into a unitary U , that is, finding U such that $V|i\rangle_A = U|i\rangle_{A'}$ for i from 1 to d_A .

- **In other words**

The recipe for realizing the POVM described by $\{F_i\}_{i=1}^n$ on a quantum state ρ is then to embed the quantum state in the Hilbert space $\mathcal{H}_{A'}$, evolve it with the unitary U , and make the projective measurement described by the PVM $\{\Pi_i\}_{i=1}^n$.

Naimark's dilation theorem (5/5)



For each POVM element, first use SVD to decompose the

Make sure every element is rank-1

- Usually the POVM element representing the inconclusive result is not rank-1, so we have to decompose it.
- For such element Π_m , we use singular value decomposition (SVD) to find rank-1 vectors $|\phi\rangle$.

$$\Pi_m = U \Sigma V^T \quad (31)$$

$$\Pi_m = \sum_i |\phi_i\rangle \langle \phi_i| \quad (32)$$

$$|\phi_i\rangle = U_i \times s_i. \quad (33)$$

Quantum circuits for isometries

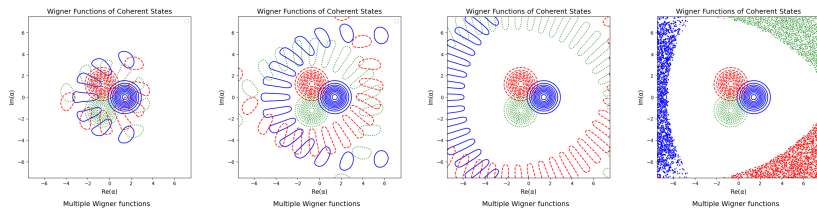
Iten et al. (2016)

Both Qiskit and qclib (Araujo et al. (2023)) implemented the three decomposition algorithms (knill, ccd, and csd) introduced in this work.

Since the size of the considered quantum states are relatively small (2 to 5 qubits), the scalability of these algorithms (noticeable for $\#qubits \geq 8$) does not affect our results.

The intuition is to estimate CNOTs of the three algorithms, then synthesize with the algorithm with the fewest CNOTs.

Example - three coherent states



(a) #qubits = 3 (b) #qubits = 4 (c) #qubits = 5 (d) #qubits = 6

Figure: Wigner functions for multiple truncated coherent states with different number of qubits. Blue, green, red represent the state $|\alpha\rangle$ with $\alpha = 1, e^{\frac{2\pi}{3}j}, e^{\frac{4\pi}{3}j}$, respectively.

Results - three coherent states

Table: Comparison of quantum circuit synthesis methods for discriminating three coherent states. Here 2Q means the number of two-qubit gates, which is also the number of CNOT gates in this case. Fidelity is rounded down to the fifth digit.

# Qubits	Fidelity	Original		Resynthesis			Approx.deg = 1.00			Approx.deg = 0.97			AQC (2Q = 10)		
		Depth	2Q	Depth	2Q	Fidelity	Depth	2Q	Fidelity	Depth	2Q	Fidelity	Depth	2Q	Fidelity
2	0.97803	84	41	45	20	1.00000	29	14	0.99999	29	14	0.59409	21	10	0.35564
3	0.99998	430	218	225	100	0.99999	83	61	0.99992	83	61	0.88925	15	10	0.96252
4	0.99999	2034	1025	1009	444	0.99999	253	252	0.99989	253	252	0.13754	11	10	0.32505
5	0.99999	8916	4474	4273	1868	0.99999	817	1020	0.99990	817	1020	0.00032	9	10	0.25978
6	0.99999	37220	18634	17585	7660	0.99999	2729	4091	0.99984	-	-	-	-	-	-

Simulation - three coherent states

- **Design under test (DUT)**

One of the predefined pure state $|\psi\rangle$ + the quantum circuit we design

- **Environment**

Increasing depolarizing noise

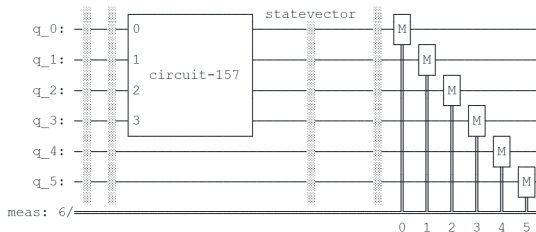
- **What to observe**

- Outcome distribution from fixed number of shots
- Probabilities from simulated state vector

We are also curious about how noise will affect the discrimination results, and we want to see if smaller circuits work well enough for the states using more qubits. We expect the circuits with high process fidelity to run successfully, so we will show the results together with noisy simulation.

David versus Goliath: 4-qubit circuit discriminating a 6-qubit state

- `sv_coh_symm_small(num_qubits=6)`
- `qc_iso_coh_q3_n3_no_backend_resynth.qasm`
- `OrderedDict('u': 208, 'cx': 100)`
- Depth: 225

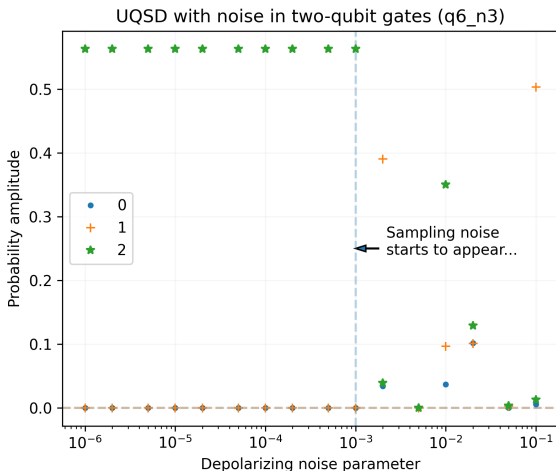


Ideal simulation

Table: Simulation results: top measurement outcomes and probabilities

Initial State	Top Outcome(s)	Counts	Probabilities
$ \psi_0\rangle$	000000	585	0.5583
	001000	261	0.2445
	001001	178	0.1971
$ \psi_1\rangle$	000001	591	0.5624
	001000	257	0.2416
	001001	176	0.1960
$ \psi_2\rangle$	000010	591	0.5627
	001000	257	0.2416
	001001	176	0.1957

Simulation with two-qubit depolarizing noise



Summary

Summary

- We have developed a tool to synthesize and test quantum circuits by specifying a quantum state discrimination problem.
- We have investigated different formulations of quantum state discrimination with mixed states and the result probability distribution in mind.

Appendix

Appendix

Coherent states

Related works

Izumi et al. (2021)

This work performed a physical experiment for UQSD for 4 coherent states.

Canonical Coherent States

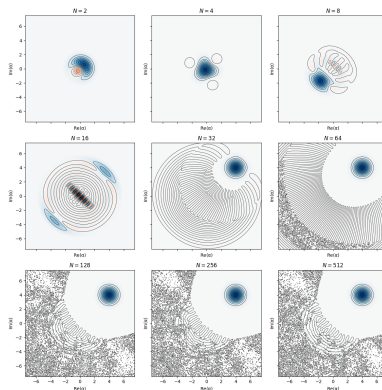
The canonical coherent states have three properties that are mutually equivalent.

- They are eigenvectors of the annihilation operator:
 $\hat{a} |\alpha\rangle = \alpha |\alpha\rangle$
- They are obtained from the vacuum by application of a unitary displacement operator: $|\alpha\rangle = (e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}) |0\rangle = D(\alpha) |0\rangle$
- They are states of minimal uncertainty: $\Delta X = \Delta P = \sqrt{\frac{\hbar}{2}}$

Example - State Selection

- Truncated coherent states and fidelity is important... We chose the following three sets of coherent states to demonstrate the dynamics of the different methods to search for a proper POVM for state discrimination.
- First, we want the set of coherent states to show some symmetric property. Since each coherent state correspond to an annihilation operator with a unique eigenvalue, we set the eigenvalues to have the same magnitude but different angles and the difference of each pair of angles is the same. We can visualize the states with Wigner function, as the figure below.

Influence of Truncation of Coherent States



Wigner Functions of Truncated Coherent States $|\alpha| = 4, \angle \alpha = \pi/4$

Influence of Truncation of Coherent States

- We also calculated the fidelity of truncated coherent states by padding zeros to the same dimension.
- A truncated coherent state with the largest dimension is selected as the reference.
- So for $\alpha = 4$, we will select $N = 64$.

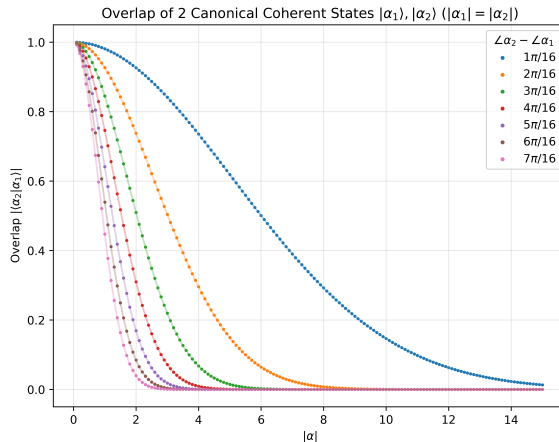
Fock States (N)		Fidelity
2		1.525e-06
4		5.193e-07
8		2.606e-04
16		2.674e-01
32		0.999594
64	1.0 (0 to 13 digits)	
128	1.0 (0 to 13 digits)	
256	1.0 (0 to 13 digits)	
512	1.0 (0 to 13 digits)	
1024	1.0 (0 to 13 digits)	

Table: Fidelity of $|\alpha\rangle$ w.r.t $N = 1024$ ($\alpha = 4$)

Selecting α

- It is also important to choose α (The eigenvalue of the annihilation operator).
- Since we know the closed form of coherent states, we can evaluate the overlap before we decide the sets of states.

Theoretical overlap of canonical coherent states



Theoretical overlap of canonical coherent states

For two different canonical coherent states $|\alpha\rangle$ and $|\beta\rangle$,

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}(|\beta|^2+|\alpha|^2-2\beta^*\alpha)} \quad (34)$$

Assume $\alpha = r_1 e^{i\theta_1}$, $\beta = r_2 e^{i\theta_2}$,

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}(r_1^2+r_2^2-2r_1r_2e^{i(\theta_1-\theta_2)})} \quad (35)$$

If $r_1 = r_2 = r$,

$$\langle\beta|\alpha\rangle = e^{-r^2(1-e^{i(\theta_1-\theta_2)})} \quad (36)$$

If $\theta_1 = \theta_2 = \theta$,

$$\langle\beta|\alpha\rangle = e^{-\frac{1}{2}(r_1-r_2)^2} \quad (37)$$

Appendix
Derivation of MED

Example (1/4)

From Equation 6, the maximal probability happens when Π_1 projects to the positive eigenspace of $p_1 |\psi_1\rangle \langle \psi_1| - p_2 |\psi_2\rangle \langle \psi_2|$. For our example,

$$\begin{aligned} p_1 |\psi_1\rangle \langle \psi_1| - p_2 |\psi_2\rangle \langle \psi_2| &= 0.5 \times \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - 0.5 \times \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\ &= 0.5 \times \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & -0.5 \end{bmatrix} \equiv 0.5A \end{aligned}$$

Example (2/4)

The eigenvectors v_1, v_2 and eigenvalues λ_1, λ_2 of A are

$$v_1 = [-1 - \sqrt{2} \quad 1]^T, \lambda_1 = \frac{\sqrt{2}}{2} \quad (38)$$

$$v_2 = [-1 + \sqrt{2} \quad 1]^T, \lambda_2 = \frac{-\sqrt{2}}{2} \quad (39)$$

We choose v_1 since we want the one with positive eigenvalue.

Example (3/4)

The resulting measurement operators are

$$\begin{aligned}
 \Pi_1 &= \frac{v_1 v_1^T}{(-1 - \sqrt{2})^2 + 1^2} = \begin{bmatrix} 3 + 2\sqrt{2} & -1 - \sqrt{2} \\ -1 - \sqrt{2} & 1 \end{bmatrix} / (4 + 2\sqrt{2}) \\
 &= (2 - \sqrt{2}) \begin{bmatrix} 3 + 2\sqrt{2} & -1 - \sqrt{2} \\ -1 - \sqrt{2} & 1 \end{bmatrix} / 4 \\
 &= \begin{bmatrix} (2 + \sqrt{2})/4 & -\sqrt{2}/4 \\ -\sqrt{2}/4 & (2 - \sqrt{2})/4 \end{bmatrix} \\
 \Pi_2 &= I - \Pi_1 = \begin{bmatrix} (2 - \sqrt{2})/4 & \sqrt{2}/4 \\ \sqrt{2}/4 & (2 + \sqrt{2})/4 \end{bmatrix}
 \end{aligned}$$

Example (4/4)

The probability of success for MED is

$$P_{succ} = p_1 \langle \psi_1 | \Pi_1 | \psi_1 \rangle + p_2 \langle \psi_2 | \Pi_2 | \psi_2 \rangle \quad (40)$$

$$= 0.5 \times (2 + \sqrt{2})/4 + 0.5 \times (2 + \sqrt{2})/4 \quad (41)$$

$$= (2 + \sqrt{2})/4 = 0.85355 \dots > 0.75 \quad (42)$$

(Return to Frame 22)

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